CHAPTER 4

Interdisciplinarity: Challenges for Research

跨学科性：研究面临的挑战

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Vast resources are available for the digital historian. As the previous discussions show, at the same time as more material is easily available, there are many heuristic and source-critical aspects that have to be taken into consideration. A major challenge is the use of large data sets. Therefore, digital history often appears as interdisciplinary work and takes the form of collaborative research based on cooperation and joint work. There is a need for transgressing disciplinary borders not only because of the complex problems awaiting resolution but also because, in other fields of research, such generic tools have been, or might have been, developed and productively deployed that can be applied in other contexts. Digital historians think about their work in terms of their interest in historical problems, but already problem-solving might require expertise from different disciplines and methodological traditions. It would be unfortunate not to take advantage of what has been found in other scientific communities and within different research paradigms.

数字历史学家可以利用极为丰富的资源。正如前文讨论所表明的，在越来越多材料可以轻易获取的同时，研究者也必须考虑许多与探索方法和史料批判有关的问题。大规模数据集的使用便是一项重大挑战。因此，数字历史往往呈现为跨学科工作，并采取以协作和共同劳动为基础的合作研究形式。我们之所以需要跨越学科边界，不仅是因为有复杂问题亟待解决，也因为其他研究领域已经开发出，或可能已经开发出一些可以有效运用并移植到其他情境中的通用工具。数字历史学家从自己对历史问题的兴趣出发理解其工作，但仅仅是解决问题这一环节，就可能需要来自不同学科和方法论传统的专长。若不利用其他科学共同体和不同研究范式中已经取得的成果，实在令人遗憾。

Digital history does not stand alone before this challenge. The world has always been complex, to be sure, but we are increasingly conscious of the global hazards we confront, which further emphasizes the need for interdisciplinary work. In his compelling book We Have Never Been Modern, originally published in French in 1991, Bruno Latour started his analysis by describing how a single edition of a newspaper describes the world around us. He observed how chemical and political reactions seemed to intertwine: in the present-day world, human and non-human factors are inseparable, a fact exemplified in Latour's observations on the threat posed by the widening of the hole in the ozone layer. Nature is not something to be externalized from human behaviour and culture.[1] The ozone question aroused fierce public debate in the 1990s, and today we can notice that it was, in the end, possible to change patterns of consumption, influence industry and decision makers and, by these means, avoid the ozone catastrophe. This positive result only confirms the fact that human and non-human agency are linked together.

数字历史并不是唯一面对这一挑战的领域。诚然，世界向来复杂，但我们日益意识到自身所面对的全球性风险，这进一步凸显了跨学科工作的必要性。布鲁诺·拉图尔那部引人入胜的著作《我们从未现代过》最初于1991年以法文出版。他在分析开篇描述了一份报纸的单期内容如何呈现我们周遭的世界。他观察到，化学反应与政治反应似乎相互交织：在当今世界，人类因素与非人类因素密不可分。拉图尔对臭氧层空洞扩大所构成威胁的观察，正是这一点的例证。自然并不是可以从人类行为与文化中外置出去的事物。[1] 20世纪90年代，臭氧问题引发了激烈的公共争论；如今我们可以看到，人们最终确实能够改变消费模式、影响产业界和决策者，并借此避免臭氧灾难。这一积极结果恰恰进一步证实，人类与非人类的行动力彼此相连。

Today, we have new challenges, such as climate change and global warming, the Earth's diminishing biodiversity and political instability that relates to these larger frameworks. From this perspective it is easy to understand the critique that Western academia has its roots in a world quite different from the present. How can we change universities so that they are simultaneously elastic and resilient to respond to these challenges while remaining firm and determined in their long-term role as cradles of serious research that foster intensive devotion to scholarly problems that widen our knowledge of the world, even in cases where these results have no obvious applicable role in the present day? While it is evident that problems like global warming cannot be solved through the means of one discipline only, it is as apparent that these problems do not mean that disciplinary thinking would be outdated.

今天，我们面临着新的挑战，例如气候变化与全球变暖、地球生物多样性的衰减，以及与这些宏大框架相关的政治不稳定。从这一角度看，我们不难理解这样的批评：西方学术界植根于一个与当下迥然不同的世界。我们该如何改变大学，使其既具备回应这些挑战所需的灵活性与韧性，同时又能坚定不移地履行其长期使命，继续作为严肃研究的摇篮，鼓励学者潜心钻研能够拓展我们对世界之认识的学术问题，即便这些成果在当下并没有显而易见的应用价值？全球变暖之类的问题显然不可能仅凭单一学科来解决；同样显而易见的是，这些问题并不意味着学科式思维已经过时。

Beyond disciplinarity

超越学科界限

The interest in overcoming disciplinary borders has certainly been part of science, the arts and humanities for a long time. There is a whole vocabulary to describe the many ways in which disciplines can entangle or intertwine. Intradisciplinary suggests a scholarly work that takes place within a single discipline only. It is disciplinary par excellence. Cross-disciplinary, in turn, refers to the effort to promote mutual understanding between disciplines. Multidisciplinary work goes further in its aim to bring researchers from different disciplinary backgrounds together to concentrate on a joint agenda so that all members can creatively draw on their specific knowledge and skills. In an interdisciplinary setting, the aim is even more ambitious, and researchers attempt to integrate knowledge and methods from various disciplines to create an overarching toolbox for the project at hand. To go even further, transdisciplinary research, which is a form of interdisciplinary work, goes a bit further in its dissatisfaction with disciplinary premises. While interdisciplinary thinking still acknowledges the role of disciplines, since it is inter, or 'in-between', transdisciplinary research tackles the challenge of going beyond disciplinary borders and perspectives to create a larger intellectual framework for research.[2] In her book Interdisciplining Digital Humanities, Julie Thompson Klein crystallized the idea in the following way: 'Transdisciplinarity transcends the narrow scope of disciplinary worldviews through an overarching synthesis with new conceptual frameworks or paradigms.'[3]

长期以来，跨越学科边界的兴趣无疑一直存在于科学、艺术与人文学科之中。人们发展出一整套词汇，用以描述学科相互牵连或交织的多种方式。学科内研究（intradisciplinary）指仅在单一学科内部开展的学术工作，是最典型不过的学科研究。交叉学科研究（cross-disciplinary）则指促进不同学科之间相互理解的努力。多学科研究（multidisciplinary）更进一步，旨在把具有不同学科背景的研究者聚集起来，围绕共同议题展开工作，使所有成员都能创造性地运用各自特有的知识与技能。在跨学科（interdisciplinary）情境中，目标更具雄心：研究者试图整合多个学科的知识和方法，为当前项目建立一套总括性的工具箱。再进一步，超学科研究（transdisciplinary）作为跨学科工作的一种形式，对学科前提持有更强的不满。跨学科思维仍然承认学科的作用，因为inter意为“在……之间”；超学科研究则直面超越学科边界与学科视角的挑战，以建立更宏大的研究思想框架。[2] 朱莉·汤普森·克莱因在《跨学科化数字人文》中将这一观念概括如下：“超学科性通过采用新的概念框架或范式进行总体综合，超越学科世界观的狭窄范围。”[3]

Based on current research and education under the rubric of 'digital history', it seems fair to argue that digital history follows mainly interdisciplinary and transdisciplinary points of departure. In interdisciplinary cases, disciplinary expertise is necessary for achieving goals, but this endeavour is only possible if disciplines unite forces. Simultaneously, and perhaps more so for the digital humanities than digital history, long-term transdisciplinary strategies are necessary for educating a generation of scholars reared without a disciplinary mindset. Both conclusions are true, since the scope, type and goals of interdisciplinary work vary. We can, no doubt, imagine a future where artificial intelligence has permeated scholarly work to such an extent that data science has become a kind of meta-science and become a substantial presence in any kind of research. In that case, data science diverges from the disciplinary register and must be integrated into researcher training on a general level. This will probably not be the case in the future, as science has always been in a state of entropy: new fields of research emerge constantly. But perhaps we need transdisciplinary centres and hubs to envisage new pathways for research.

根据目前归入“数字历史”名下的研究与教育，我们有理由认为，数字历史主要以跨学科和超学科作为出发点。在跨学科案例中，实现目标离不开学科专长，但只有当不同学科联合起来，这一事业才有可能完成。与此同时，培养一代不受学科定势束缚的学者，需要长期的超学科策略；这一点对数字人文而言，或许比对数字历史更加重要。这两项判断都成立，因为跨学科工作的范围、类型和目标各不相同。我们当然可以设想这样一种未来：人工智能广泛渗入学术工作，以至于数据科学成为某种元科学，并在任何类型的研究中占据重要位置。在这种情况下，数据科学将脱离既有的学科谱系，必须在一般层面上纳入研究者训练。不过，未来大概不会完全如此，因为科学一直处于熵增状态，新的研究领域不断涌现。但或许，我们确实需要超学科中心和枢纽，以构想新的研究路径。

In the end, everything depends on how the scope, type and goals of interdisciplinarity have been defined. These three dimensions were itemized by Katri Huutoniemi, Julie Thompson Klein, Henrik Bruun and Janne Hukkinen in their 2010 analysis on how interdisciplinarity was articulated in a selection of concrete research projects.[4] For them, the scope refers to the question of what is actually integrated into a particular project in which disciplines meet. Interdisciplinarity in a project is seldom comprehensive but is narrowed down to a few areas. The second dimension is the type of interdisciplinarity, which means paying attention to how the research will finally be realized, 'not only in the framing of research problems and coordinating knowledge flows between fields, but also in the execution of research and the formulation and analysis of results'.[5] The third dimension emphasizes the importance of the actual goals of the interdisciplinary enterprise. Why does interdisciplinarity take place and how do researchers articulate these goals?[6]

归根结底，一切都取决于人们如何界定跨学科性的范围、类型与目标。卡特里·胡托涅米、朱莉·汤普森·克莱因、亨里克·布鲁恩和扬内·胡基宁在2010年的一项分析中，考察了若干具体研究项目如何表述跨学科性，并将其分为上述三个维度。[4] 在他们看来，“范围”涉及的问题是：在一个学科相遇的具体项目中，究竟整合了哪些内容。一个项目中的跨学科性很少是全面的，通常只会收窄到少数几个领域。第二个维度是跨学科性的“类型”，这意味着要关注研究最终如何落实，“不仅包括研究问题的框定和不同领域之间知识流动的协调，也包括研究的实施以及结果的形成与分析”。[5]
第三个维度强调跨学科事业实际目标的重要性：跨学科为何发生？研究者又如何表述这些目标？[6]

In their article, Huutoniemi et al. maintain that these goals for interdisciplinary work are different in nature: some projects spring out of an interest in gaining more in-depth knowledge, while others have a more instrumental and perhaps non-academic agenda. This is a characteristic of any research, not only digital humanities, but the article makes an important point in acknowledging that these goals and their articulations can be analysed and evaluated. In the case of the digital humanities, however, there is a wider aspect of science policy that should be kept in mind and that goes beyond the question of goals. Most research activities are directed by funding bodies, by research programmes and university strategies, both nationally and internationally. This can be seen as a threat for the freedom of research. What happens if researchers must justify their work through engagement with other fields, particularly in a situation where the funding of more disciplinary-based research is diminishing? Yu-wei Lin expressed this in 2012, highlighting the fact that 'national and international research funding councils have increasingly emphasized that research in humanities should engage with data-intensive and evidence-based academic activities, as those in natural sciences and engineering do'.[7] This concern is understandable, and there might be some truth in it. Still, it is difficult to find evidence that interdisciplinarity or engagement with data-intensive fields of research automatically translates into a loss for humanities. Interdisciplinarity is never a one-way street. It is a process of mutual exchange in which the humanities' viewpoints may also impact researchers from other disciplines. As the world today is saturated with information technology and digital devices, it is even more urgent that historians and other researchers in the humanities engage with research in such fields as data science and engineering.

胡托涅米等人在文章中指出，跨学科工作的这些目标在性质上并不相同：有些项目源于获取更深入知识的兴趣，另一些项目则带有更强的工具性议程，甚至可能是非学术性的。这是任何研究都具有的特征，并非数字人文所独有；但该文提出了一个重要观点，即这些目标及其表达方式可以被分析和评估。不过，就数字人文而言，还应关注一个超出目标问题之外、更广泛的科学政策层面。无论在国内还是国际上，大多数研究活动都受到资助机构、研究计划和大学战略的引导。这可能被视为对研究自由的威胁。如果研究者必须通过与其他领域合作来证明自身工作的合理性，尤其是在更具学科本位的研究经费不断减少的情况下，会发生什么？林育玮在2012年表达了这一忧虑，并特别指出：“国家和国际研究资助委员会日益强调，人文学科研究应像自然科学和工程学那样，参与数据密集型和证据导向的学术活动。”[7] 这一担忧可以理解，也可能不无道理。不过，我们很难找到证据证明，跨学科性或参与数据密集型研究领域必然会给人文学科造成损失。跨学科从来不是单行道，而是一个相互交流的过程；在此过程中，人文学科的观点也可能影响其他学科的研究者。当今世界充斥着信息技术和数字设备，历史学家及其他人文学科研究者更有迫切需要参与数据科学、工程学等领域的研究。

Trading across borders

跨越边界的交换

Interdisciplinarity is not a simple counting operation from which the result is only the sum of its parts. In the most fruitful and rewarding cases, it leads to observations that would otherwise not have been possible. Often in research, the findings of one discipline are applicable in another disciplinary context, where they can lead to a breakthrough. It is possible, and constructive, to experiment with methods and research strategies over disciplinary borders. An interesting example is a project in organic chemistry conducted by researchers at the Pedagogical University in Krakow, the Polish Academy of Sciences and the Ulsan National Institute of Science and Technology in South Korea. The point of departure was the major problem of the 'enormous number of different molecules and their fragments that play a role in chemical reactions'. The aim was to try to understand why certain clusters of atoms tend to keep together while repulsing some other groups. The researchers decided to compare chemical units using language tools and split the complex molecules into 'meaningful' substructures. After that, they applied text mining tools, especially topic modelling, in clustering these substructures. This was only an experiment, but in the end, they were able to conclude that semantic information can be 'extracted from chemical corpora using text mining algorithms'.[8] In this case, the tools developed in the digital humanities could be employed in solving problems in natural sciences. It is only a matter of imagination and the courage to experiment with the idea.

跨学科性并不是简单的加法运算，其结果绝非各部分之和而已。在最富成效、最有回报的案例中，它能够带来原本不可能获得的观察。研究中，一个学科的发现往往可以应用于另一个学科情境，并在那里促成突破。跨越学科边界试验不同的方法和研究策略，既是可行的，也是建设性的。一个有趣的例子是克拉科夫教育大学、波兰科学院和韩国蔚山科学技术院的研究者共同开展的一项有机化学项目。该项目的出发点是一个重大难题：参与化学反应的“ 不同分子及其片段数量极其庞大 ”。研究旨在理解，为何某些原子簇倾向于聚合在一起，同时又排斥其他原子团。研究者决定使用语言工具比较化学单元，并把复杂分子拆分成“ 有意义的 ”子结构。随后，他们应用文本挖掘工具，尤其是主题建模，对这些子结构进行聚类。这起初只是一次试验，但他们最终得出结论：可以“ 利用文本挖掘算法从化学语料库中提取 ”语义信息。[8] 在这个案例中，数字人文所开发的工具可以用来解决自然科学中的问题。所需要的，只是想象力以及试验这一构想的勇气。

Another example of how different fields of research can benefit each other, even fields that at first glance do not seem to have much in common, is the software BLAST (Basic Local Alignment Search Tool), which was developed at the National Center for Biotechnology Information. BLAST finds areas of similarity between biological sequences and can be used to compare, for example, nucleotide and protein sequences. It is a software that all researchers within bioinformatics know. The software itself is, however, generic by nature: it detects similar sequences and can be applied to other kinds of sequences, not necessarily biological. In a recent project at the University of Turku, BLAST was applied in the study of text reuse in old newspapers, which include many errors in optically recognized text, mainly because nineteenth-century papers in Finland used Gothic type, which is difficult for OCR software to process. BLAST is built to be tolerant of errors and mutations and might thus be useful for identifying similarities in messy material. In the project, each character was encoded as an amino acid, and then the whole corpus was run through BLAST. In the end, 61 million occurrences of similarity were detected from a corpus of 5.1 million pages, and these occurrences were clustered into repetition chains.[9] The result was a text reuse database that other researchers can query.[10] This unexpected methodological solution would not have been possible if the research team did not include experts in bioinformatics, but its foundation was the fact that there was an interdisciplinary research group in the first place. Furthermore, it must be added that methods and research questions are always interrelated. It is possible to first identify a problem that needs an interdisciplinary solution, but it is equally important to note that interdisciplinary practices stimulate the team's imagination and help locate new and fruitful research questions, problems that might otherwise have gone unnoticed.

另一个能够说明不同研究领域如何彼此受益的例子，是由美国国家生物技术信息中心开发的BLAST软件（基本局部比对搜索工具）；即使某些领域乍看之下似乎没有多少共同点，也可以从中相互获益。BLAST能够找出生物序列之间的相似区域，例如可用于比较核苷酸序列和蛋白质序列。它是所有生物信息学研究者都熟悉的软件。不过，该软件本质上具有通用性：它检测相似序列，也可以应用于其他类型的序列，而不一定是生物序列。图尔库大学最近的一个项目把BLAST用于研究旧报纸中的文本复用。这些报纸经光学字符识别得到的文本含有大量错误，主要因为19世纪芬兰报纸使用哥特字体，而OCR软件很难处理这种字体。BLAST在设计上能够容忍错误和突变，因此可能有助于识别杂乱材料中的相似之处。项目中，每个字符都被编码为一种氨基酸，然后研究者用BLAST处理整个语料库。最终，他们从510万页的语料库中检测到6100万处相似实例，并将这些实例聚类成重复链。[9]由此形成了一个可供其他研究者查询的文本复用数据库。[10] 如果研究团队中没有生物信息学专家，这一出人意料的方法论方案就不可能实现；而它的基础，首先就在于项目从一开始便拥有一个跨学科研究团队。此外还必须指出，方法与研究问题始终相互关联。我们固然可以先发现一个需要跨学科方案的问题，但同样重要的是认识到，跨学科实践会激发团队的想象力，帮助他们找到新颖而富有成效的研究问题，包括那些原本可能被忽视的问题。

Interdisciplinary work often needs shared spaces. Patrik Svensson has written extensively on the role of spaces in the digital humanities, both concrete locations and abstract spatialities, which is inspiring also for digital history. He has described the digital humanities as a trading zone and a meeting place where people, ideas, tools and methods can mingle.[11] In this sense, the application of BLAST was an outcome of a practice of 'trading' and 'meeting'. However, cooperation is not always so harmonious. Svensson refers to Mary Louise Pratt's notion of the contact zone. According to Pratt, cultures often meet 'in highly asymmetrical relations of power'. The contact zone can be imbalanced and hierarchical as a platform for communication and does not necessarily lead to trade or mutual exchange.[12] This could happen, for example, in cases when one of the partners does not recognize the premises of the other or if there is no willingness to negotiate on the basic concepts. Max Kemman has pointed out that quite often there is 'knowledge asymmetry in interdisciplinary collaboration'. He refers to a project in which 'historians were unaware how the computer scientists would perform their tasks, and as a result lacked the power to properly negotiate a tool that would work to their satisfaction'.[13] Kemman has discussed this problem further, and in more detail, in his 2019 study Trading Zones of Digital History. He based his research on wide-ranging source material, including an online survey, interviews, his own observations and the LDA topic modelling of over 10,000 blog posts by historians between 2008 and 2017. In his conclusion, he noted that there was a feeling of uncertainty in digital history projects, arising on the one hand from historians' hesitation on how to use computational methods, and on the other from data scientists' uncertainty on how their methodological toolbox fits with historical data sets. Kemman's analysis revealed different ways of solving these problems: some historians, which he called 'digital history brokers', were able to use the trading zone as a creative space and negotiate with different groups, including policymakers. On a macro level, however, the solutions of the trading zone had not yet changed research practices.[14]

跨学科工作往往需要共享空间。帕特里克·斯文松曾广泛论述空间在数字人文中的作用，其中既包括具体场所，也包括抽象空间性；这些论述对数字历史同样具有启发。他把数字人文描述为一个“交易区”和会面场所，人、思想、工具与方法可以在其中交汇。[11] 从这个意义上说，BLAST的应用正是“交易”与“会面”实践的产物。然而，合作并不总是如此和谐。斯文松援引了玛丽·路易丝·普拉特的“接触区”概念。按照普拉特的说法，文化常常“在高度不对称的权力关系中”相遇。作为沟通平台，接触区可能失衡并具有等级性，未必会带来交易或相互交流。[12] 例如，当一方不承认另一方的前提，或双方不愿就基本概念展开协商时，就可能出现这种情况。马克斯·凯曼指出，“跨学科合作中经常存在知识不对称”。他提到一个项目，其中“历史学家并不了解计算机科学家将如何执行任务，结果便没有足够权力去充分协商，以获得一种令他们满意的工具”。[13] 凯曼在2019年的研究《数字历史的交易区》中进一步、更详细地讨论了这个问题。他的研究建立在广泛的材料之上，包括在线调查、访谈、他本人的观察，以及对历史学家在2008至2017年间发布的逾一万篇博客文章进行LDA主题建模。他在结论中指出，数字历史项目中存在一种不确定感：一方面源于历史学家对如何使用计算方法犹豫不决，另一方面源于数据科学家不确定自己的方法论工具箱如何适配历史数据集。凯曼的分析揭示了不同的解决方式：一些被他称为“数字历史经纪人”的历史学家，能够把交易区用作创造性空间，并与包括政策制定者在内的不同群体展开协商。然而在宏观层面，交易区中的这些解决方案尚未改变研究实践。[14]

Towards interdisciplinary practices

迈向跨学科实践

The question of uncertainty is critical in any interdisciplinary exploration. One of the reasons for this feeling is without a doubt the tension between disciplinary thinking and interdisciplinary pressure. As Yu-wei Lin pointed out, academics come under significant pressure to break boundaries, and there is even designated funding for these kinds of projects. Nevertheless, researchers' enthusiasm and commitment to their work can be disciplinary by background: they have come to the field to devote themselves completely to it and have refined their skills specifically from that perspective. Therefore, they might be outside their comfort zone in other areas. Many key concepts in research have a disciplinary linkage, and this in itself has value. We have only to note how profoundly such research fields as cultural and gender studies have influenced humanities since the 1980s and 1990s. It has been fruitful to dedicate oneself to developing, for example, the concept of culture and gender. This is only to say that excellent research can be conducted both from a disciplinary and interdisciplinary point of departure.

在任何跨学科探索中，不确定性都是一个关键问题。这种感受无疑部分源于学科式思维与跨学科压力之间的张力。正如林育玮所指出的，学者面临着打破边界的巨大压力，甚至还有专门为这类项目设置的经费。然而，研究者对工作的热情与投入可能植根于特定学科：他们进入某一领域，全身心投身其中，并专门从该领域的视角磨炼技能。因此，到了其他领域，他们可能会离开自己的舒适区。研究中的许多关键概念都与特定学科相连，而这种联系本身具有价值。我们只需注意到，自20世纪80和90年代以来，文化研究、性别研究等领域对人文学科产生了何等深刻的影响。专心发展诸如文化和性别等概念，已经结出丰硕成果。这里要说的只是：无论以学科为出发点，还是以跨学科为出发点，都可以开展卓越的研究。

Digital history, however, is not a discipline; it is a branch of historical practice that can also be an interdisciplinary enterprise in which the researcher must enter the trading zone. This entrance might raise questions about one's academic identity or at least emphasize the skills and knowledge one has gained over the years. This leads to several questions. Where are the boundaries of one's own skills and knowledge? To what extent is it possible to develop one's own expertise to better match the interdisciplinary challenge, and how does one integrate into broader collaborations, especially if one comes from a scholarly background in which research has traditionally been a solitary enterprise? It is natural to feel uncertain in the face of such questions, but the situation is the same for all involved. One possible strategy for opening doors to interdisciplinary exchange is to try to become more conscious of those transparent, often unarticulated practices that all fields of research contain. Very few historians really explicate how they work in practice with their sources. What is the first procedure they undertake in an archive or when they open a book? The same can be asked of a data scientist. What are the basic practices with a data set? Successful interdisciplinary work must identify what those unarticulated procedures are. These habitual processes have been carried on by tradition and nurtured by academic curricula, and seldom are they encapsulated in a few lines. If we understand how we work, we can better understand where the joint undertaking with big data and other digital resources can lead us.

然而，数字历史并不是一门学科，而是历史实践的一个分支；它也可以成为一项跨学科事业，研究者必须在其中进入“交易区”。进入这一空间可能会引发有关个人学术身份的问题，或至少会凸显一个人多年来积累的技能与知识。由此产生若干问题：一个人自身技能与知识的边界在哪里？我们能在多大程度上发展自己的专长，使其更好地应对跨学科挑战？又该如何融入更广泛的合作，尤其当一个人来自某种传统上把研究视为独自事业的学术背景时？面对这些问题，感到不确定是自然的，但所有参与者的处境都一样。为跨学科交流打开大门，一种可行策略是更有意识地审视所有研究领域中那些看似透明、因而往往未被言明的实践。很少有历史学家真正明确说明，自己在实践中究竟如何处理史料。他们走进档案馆，或打开一本书时，首先会做什么？我们也可以向数据科学家提出同样的问题：面对一个数据集时，最基本的操作是什么？成功的跨学科工作必须辨识这些未被言明的程序。这些习惯性过程由传统延续，并经由学术课程培养，很少能被浓缩成寥寥数行。如果我们理解自己的工作方式，就能更好地理解，与大数据及其他数字资源共同开展的事业能够把我们带向何方。